closestAngle: real

closestDistance: real

proposal: NavigationProposal

OAApplied

Avoidance

Initialise

entry move(0.0, 0.0)

WaitProposal

proposal?requested

CheckArrival

[not requested.reached]

[OA_done]/move(av, lv); OAApplied

[requested.reached]

Arrived

entry move(0.0, 0.0); OAApplied

F

DetermineObstacle

entry OA_done = false

SampleAngle

closestAngle?closest_angle

SampleDistance

closestDistance?closest_distance

SelectCommand

[closest_distance>=min_range/\closest_distance<max_range/\closest_angle>=-90.0/\closest_angle<=90.0]/current_speed = requested.linear

[closest_angle>0.0]/av = (closest_angle-100.0)*pi/180.0

TurnAway

[closest_angle<=0.0]/av = (closest_angle+100.0)*pi/180.0

SlowDown

[closest_distance<min_range\/closest_distance>=max_range\/closest_angle<-90.0\/closest_angle>90.0]/av = requested.angular; lv = requested.linear; OA_done = true

[closest_distance>0.4]/lv = current_speed/2.0

[closest_distance<=0.4]/lv = 0.0

CloseAhead

[closest_distance<0.4/\closest_angle>-30.0/\closest_angle<30.0]/lv = -0.4; OA_done = true

F

[closest_distance>=0.4\/closest_angle<=-30.0\/closest_angle>=30.0]/OA_done = true